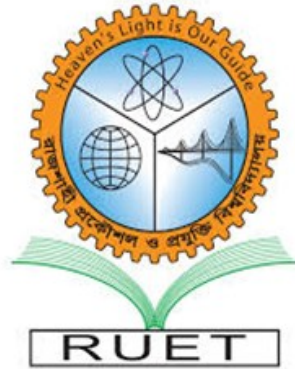

Heaven's Light is Our Guide



Rajshahi University of Engineering & Technology (RUET)

Department of Electrical and Computer Engineering

Lab report-02

Course title: Digital Techniques Sessional

Course Code: ECE 2112

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Problem-01: Simplification and Logic Circuit Implementation of the Boolean Expression $F(A, B, C): \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$.

Objective:

- To simplify the given Boolean expression using Boolean algebra laws.
- To verify the simplified expression using a truth table.
- To design and implement the corresponding logic circuit using logic gates.
- To compare the original and simplified Boolean expressions and verify that they produce the same output.

Theory:

Boolean algebra is a mathematical system used to analyze and simplify digital logic circuits. In Boolean algebra, variables can have only two possible values: 0 (LOW) and 1 (HIGH).

The basic Boolean operations are:

- AND operation ($()$) — represented by multiplication.
- OR operation ($+$) — represented by addition.
- NOT operation ($()$) — represents the complement of a variable.

The given Boolean function is:

$$F(A, B, C): \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

Here, A, B, and C are Boolean input variables, while F is the output.

Boolean algebra laws can be used to reduce the number of terms and logic gates required to implement a circuit. A simplified Boolean expression generally results in a simpler, more efficient, and less hardware-intensive logic circuit.

Simplification:

$$\begin{aligned} F(A, B, C): & \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC \\ = & \bar{A}BC + ABC + A\bar{B}C + ABC + AB\bar{C} + ABC \\ = & AB + BC + CA \end{aligned}$$

Circuit Diagram:

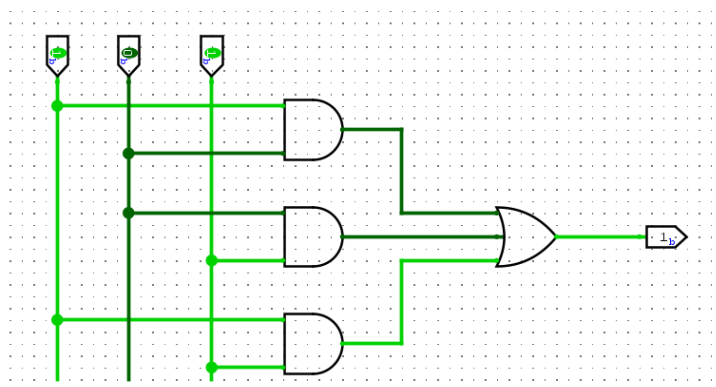


Figure 1: Logic circuit

Discussion:

In this experiment, the given Boolean expression $F(A,B,C) = \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$ was simplified using Boolean algebra. The expression represents a three-variable logic function in which the output becomes 1 for the input combinations 011, 101, 110, and 111. After applying the Boolean algebra laws, the expression can be simplified to $F = AB + AC + BC$. The truth table of the original and simplified expressions was also analyzed, and both expressions produced the same output for all possible input combinations. Thus, simplification successfully reduces the complexity of the logic circuit without changing its logical behavior.

Conclusion:

The given Boolean expression $F(A,B,C) = \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$ was successfully simplified to $F = AB + AC + BC$ using Boolean algebra. The truth table verified that the original and simplified expressions are logically equivalent. The simplified expression requires fewer logic gates, resulting in a simpler and more efficient logic circuit. Therefore, Boolean algebra is an effective method for reducing circuit complexity while maintaining the required logical functionality.

Problem-02: Simplification and Logic Circuit Implementation of the Boolean Expression $F(A, B, C): A(A+B)(A+B+C)$.

Objective:

- To simplify the given Boolean expression using Boolean algebra laws.
- To understand and apply different Boolean identities and laws during simplification.
- To determine the simplified form of the given Boolean expression.
- To design and implement the corresponding logic circuit using logic gates.
- To verify that the original and simplified expressions are logically equivalent.

Theory:

Boolean algebra is a mathematical system used to analyze and simplify digital logic circuits. In Boolean algebra, variables can have only two possible values: 0 (LOW) and 1 (HIGH).

The basic Boolean operations are:

- AND operation ($()$) — represented by multiplication.
- OR operation ($+$) — represented by addition.
- NOT operation ($()$) — represents the complement of a variable.

The given Boolean function is:

$$F(A, B, C): A(A+B)(A+B+C)$$

Here, A, B, and C are Boolean input variables, while F is the output.

Boolean algebra laws can be used to reduce the number of terms and logic gates required to implement a circuit. A simplified Boolean expression generally results in a simpler, more efficient, and less hardware-intensive logic circuit.

Simplification:

$$F(A, B, C): A(A+B)(A+B+C)$$

$$= (A+AB)(A+B+C)$$

$$= A(B+1)(A+B+C)$$

$$= A(A+B+C)$$

$$= A+AB+AC$$

$$= A(1+B+C)$$

$$= A$$

Circuit Diagram:



Figure 2: Logic circuit

Discussion:

In this experiment, the given Boolean expression $F(A,B,C)=A(A+B)(A+B+C)$ was simplified using Boolean algebra. By applying the Absorption Law successively, the expression was reduced to $F=A$. This shows that the variables B and C are redundant and do not affect the final output. The simplification significantly reduces the complexity of the logic circuit because the original expression requires multiple logic operations, whereas the simplified expression requires only a direct connection between input A and output F. Therefore, Boolean algebra provides an effective method for minimizing logic circuits and reducing unnecessary hardware.

Conclusion:

The given Boolean expression $F(A,B,C)=A(A+B)(A+B+C)$ was successfully simplified using Boolean algebra to obtain $F=A$. The simplification demonstrates that the output is completely dependent on input A, while inputs B and C have no influence on the final result. Consequently, the corresponding logic circuit can be implemented simply by connecting A directly to F. Thus, the experiment successfully demonstrates how Boolean algebra can be used to simplify expressions and minimize the complexity of digital logic circuits.

Problem-03: Simplification and Logic Circuit Implementation of the Boolean Expression $F(A, B, C): (A + \overline{BC})(AB+ABC)$.

Objective:

- To simplify the given Boolean expression using Boolean algebra laws.
- To apply De Morgan's Law, Complement Law, and Absorption Law for Boolean simplification.
- To determine the simplified form of the given Boolean function.
- To understand the relationship between Boolean expressions and their corresponding logic circuits.
- To verify that the simplified expression produces the same logical output as the original expression.

Theory:

Boolean algebra is a mathematical system used to analyze, simplify, and implement digital logic circuits. Boolean variables can have only two possible values: 0 (LOW) and 1 (HIGH).

The basic operations of Boolean algebra are:

- AND () — represented by multiplication.
- OR (+) — represented by addition.
- NOT () — represents the complement of a variable.

Some important Boolean laws used in this experiment are:

De Morgan's Laws

$$\overline{(A + B)} = \overline{A} \overline{B}$$

$$\overline{(\overline{A} \overline{B})} = A + B$$

Complement Law

$$A \overline{A} = 0$$

$$A + \overline{A} = 1$$

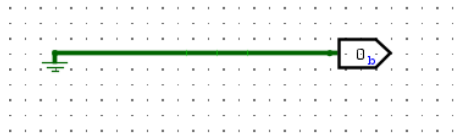
Absorption Law

$$A + AB = A$$

Boolean simplification helps reduce unnecessary logic operations and minimizes the number of gates required to implement a digital circuit.

Simplification:

$$\begin{aligned}
 F(A, B, C) &: \overline{(A + \overline{BC})}(AB + ABC) \\
 &= (\overline{A} \cdot \overline{\overline{BC}}) \cdot AB(1 + C) \\
 &= (\overline{A}BC) \cdot AB \\
 &= 0
 \end{aligned}$$

Circuit Diagram:**Figure 3: Logic circuit****Discussion:**

In this experiment, the given Boolean function was simplified using several Boolean algebra laws, including De Morgan's Law, Absorption Law, and Complement Law. The simplification showed that the output is always LOW, regardless of the values of the input variables. As a result, the logic circuit can be greatly simplified, eliminating the need for conventional logic gates. This demonstrates how Boolean algebra can effectively reduce circuit complexity, minimize hardware requirements, and simplify digital circuit implementation.

Conclusion:

The Boolean function was successfully simplified using appropriate Boolean algebra laws. The final result showed that the output remains LOW for all possible combinations of the input variables. Therefore, the simplified circuit can be implemented using a constant LOW logic connection. The experiment demonstrates the importance of Boolean algebra in minimizing logic circuits while maintaining their required functionality.

Problem-04: Simplification and Logic Circuit Implementation of the Boolean Expression $F(A, B, C): (\overline{B}(A+B)+(A+B)(A+\overline{B}))\overline{B}$.

Objective:

- To simplify the given Boolean expression using Boolean algebra laws.
- To apply Distributive, Complement, Identity, Absorption, and Factoring laws for simplification.
- To obtain the simplest form of the given Boolean function.
- To design the corresponding logic circuit from the simplified expression.
- To understand how Boolean simplification can reduce the complexity of a digital logic circuit.

Theory:

Boolean algebra is a mathematical system used to analyze, simplify, and implement digital logic circuits. Boolean variables can have only two possible values: 0 and 1.

The three basic Boolean operations are:

- AND operation ($()$) — represented by multiplication.
- OR operation ($+$) — represented by addition.
- NOT operation ($()$) — represents the complement of a variable.

Some important Boolean laws used in this experiment are:

Complement Law

$$A\overline{A}=0$$

$$A+\overline{A}=1$$

Distributive Law

$$A(B+C)=AB+AC$$

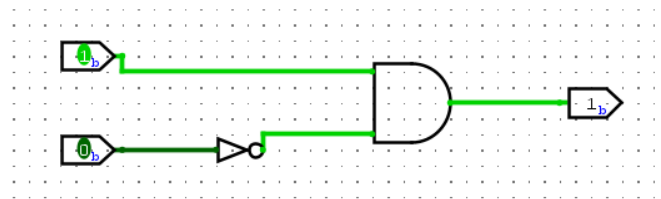
Absorption Law

$$A+AB=A$$

Boolean simplification is important because it can reduce the number of logic gates and connections required to implement a circuit.

Simplification:

$$\begin{aligned}
F(A, B, C): & (\bar{B}(A+B) + (A+B)(A+\bar{B}))\bar{B} \\
& = (\bar{A}\bar{B} + A\bar{B} + AB + A\bar{B})\bar{B} \\
& = (A + AB + A\bar{B})\bar{B} \\
& = A\bar{B}(1 + B + \bar{B}) \\
& = A\bar{B}
\end{aligned}$$

Circuit Diagram:**Figure 4: Logic circuit****Discussion:**

In this experiment, the given Boolean function was simplified using Boolean algebra laws. The common factor was first taken out to reduce the number of terms, followed by the application of the Idempotent, Complement, Identity, and Distributive laws. The expression was ultimately reduced to a simple AND operation between A and the complement of B. The input C does not appear in the given function and therefore has no effect on the output. The simplified circuit requires only one NOT gate and one AND gate, making it considerably simpler than a direct implementation of the original expression.

Conclusion:

The given Boolean function was successfully simplified using Boolean algebra techniques. The final simplified form was found to be $A\bar{B}$. This indicates that the output becomes HIGH only when A is HIGH and B is LOW, while the value of C has no effect on the output. The corresponding logic circuit can be implemented using only one NOT gate and one AND gate. Thus, the experiment demonstrates how Boolean algebra can effectively reduce logic circuit complexity and hardware requirements.